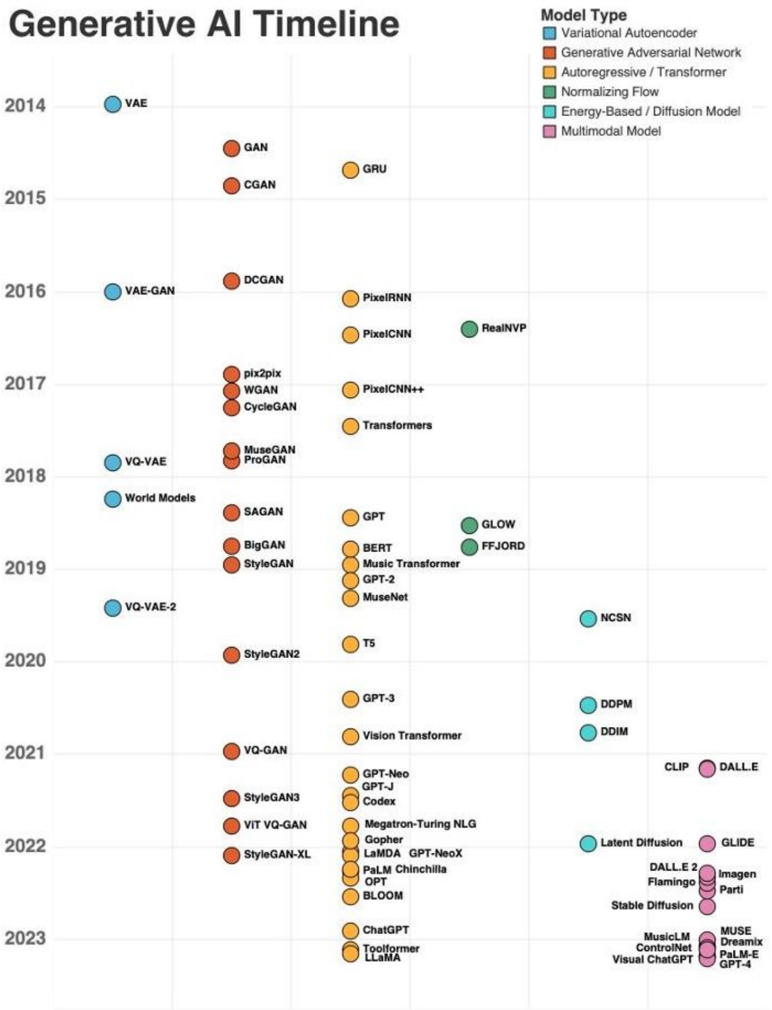
Генеративные модели (Автокодировщики + GAN)

Марк Блуменау

Thanks to Elena Kantonistova & Ilia Nikitin

О чем речь?

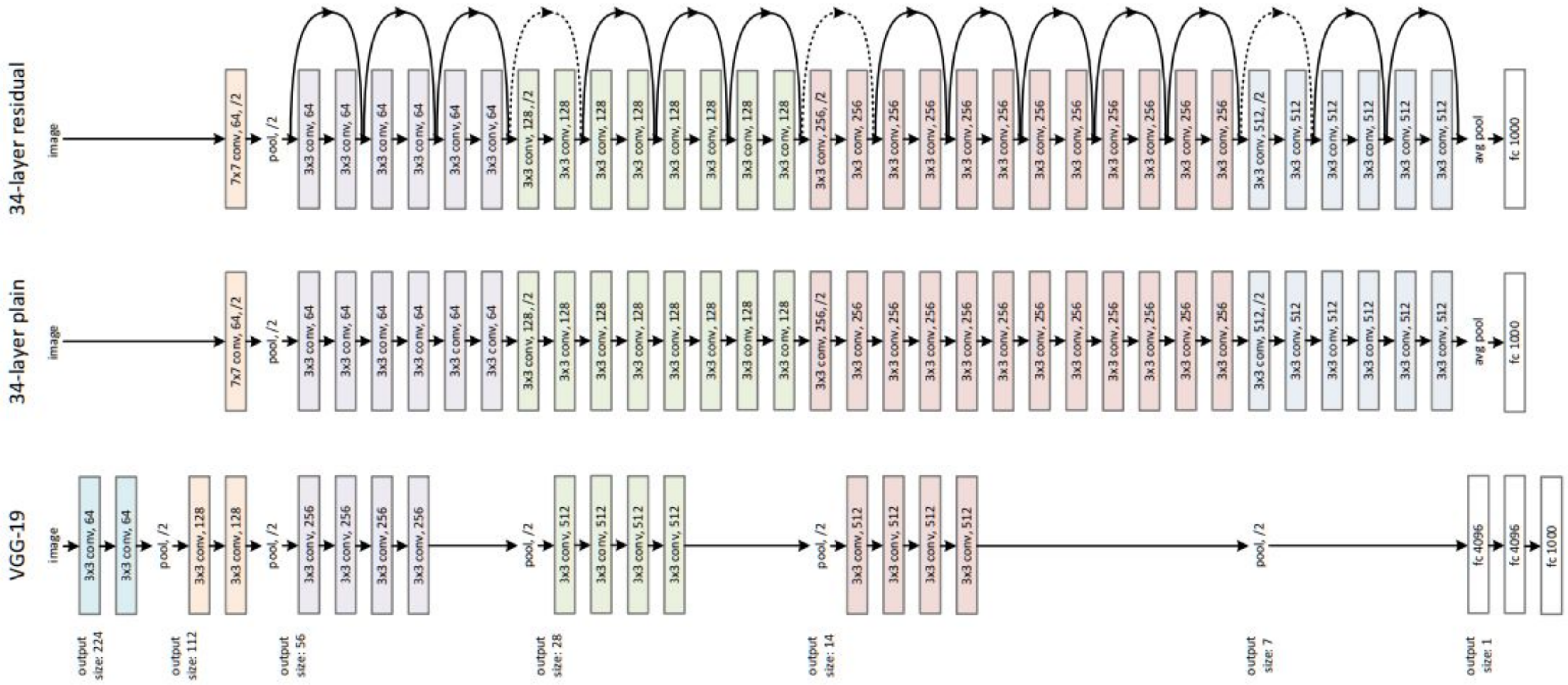


https://www.kaggle.com/discussions/general/399686

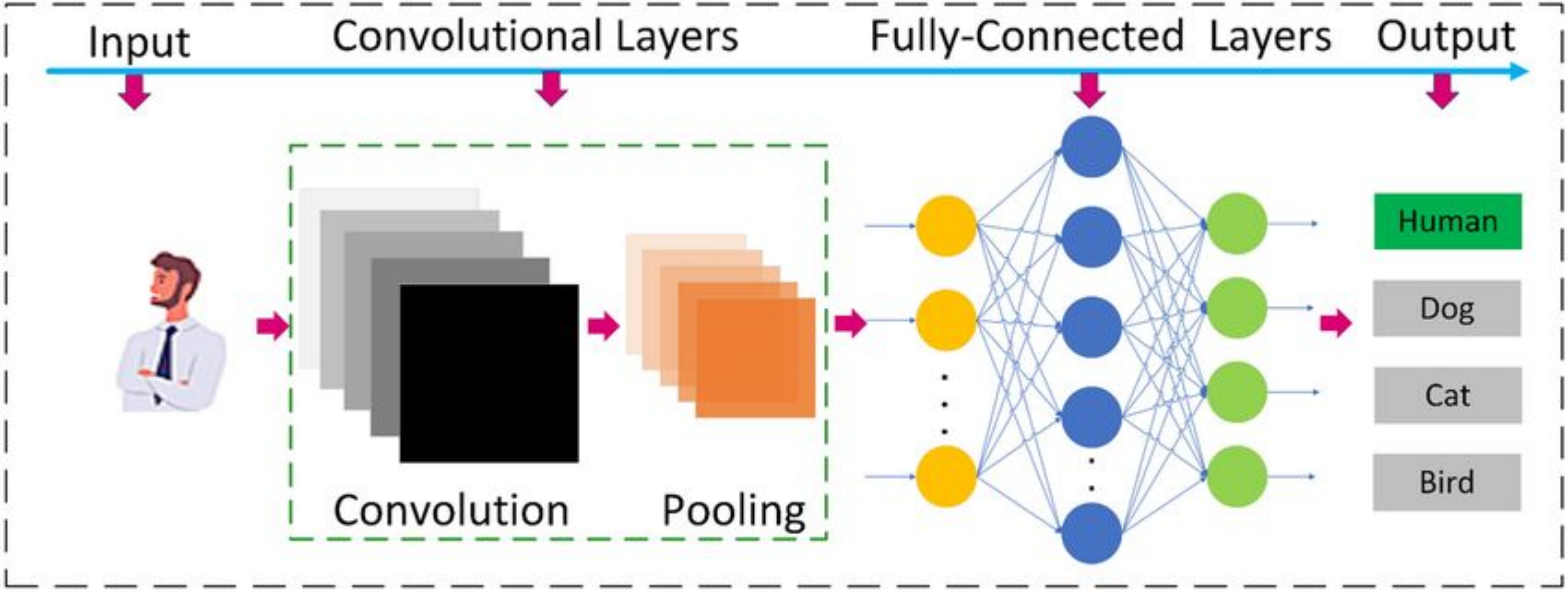
Credit: Elena Kantonistova

[Residual Networks (ResNet) - Deep Learning - GeeksforGeeks](#)

Что мы уже знаем



Supervised embeddings



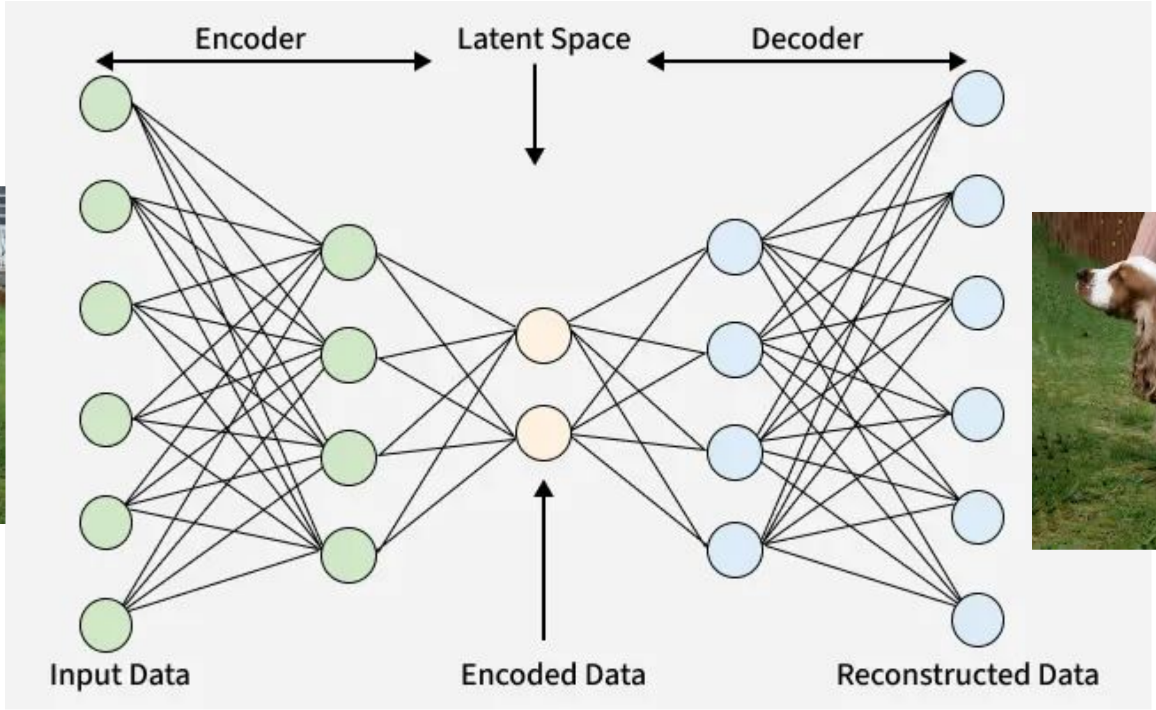
https://www.researchgate.net/figure/The-complete-internal-structure-of-the-CNN-model-for-image-classification_fig4_376619384

А если разметки нет?

То смотрим старую лекцию?

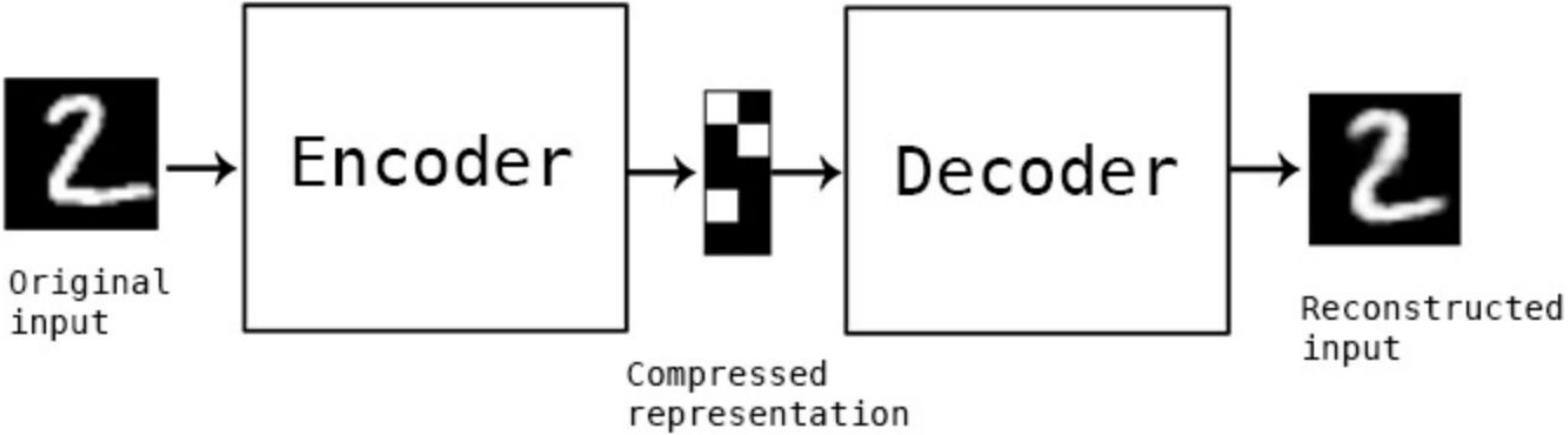
[Types of Autoencoders - GeeksforGeeks]

Нет!



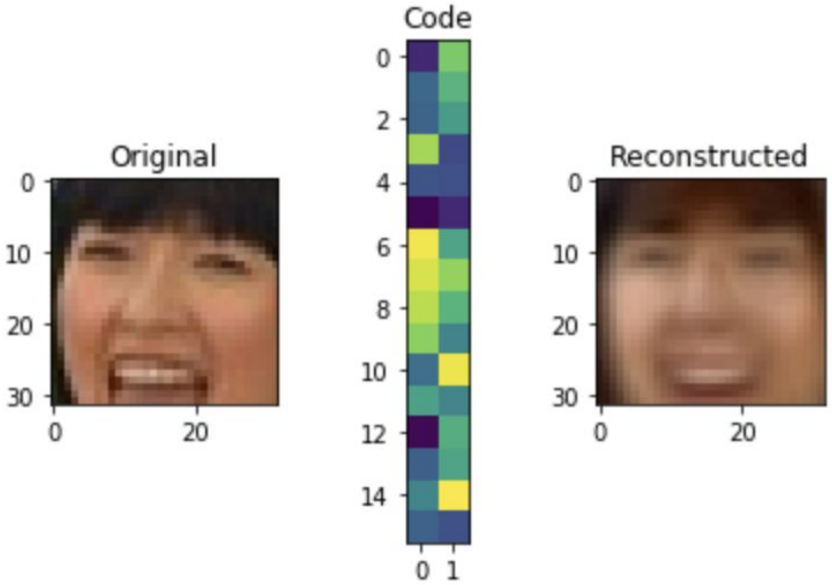
Как учить?

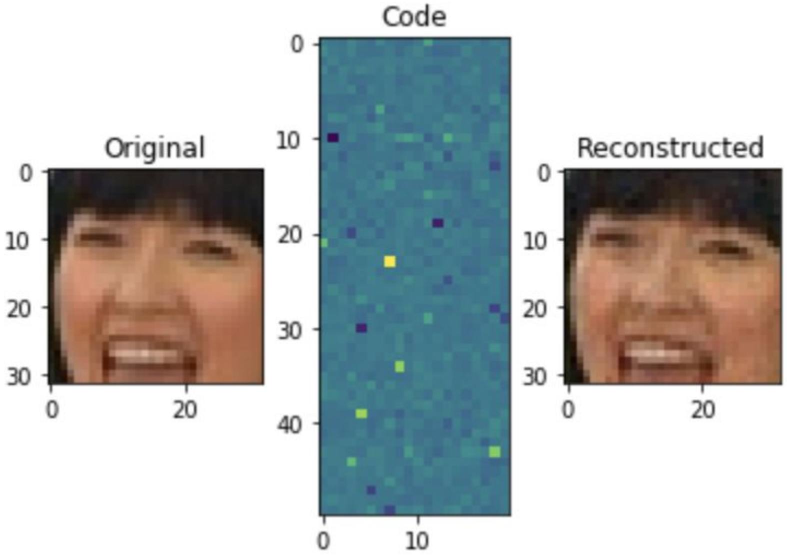
$$L\Big(x_i, g\big(f(x_i)\big)\Big) \to \min$$



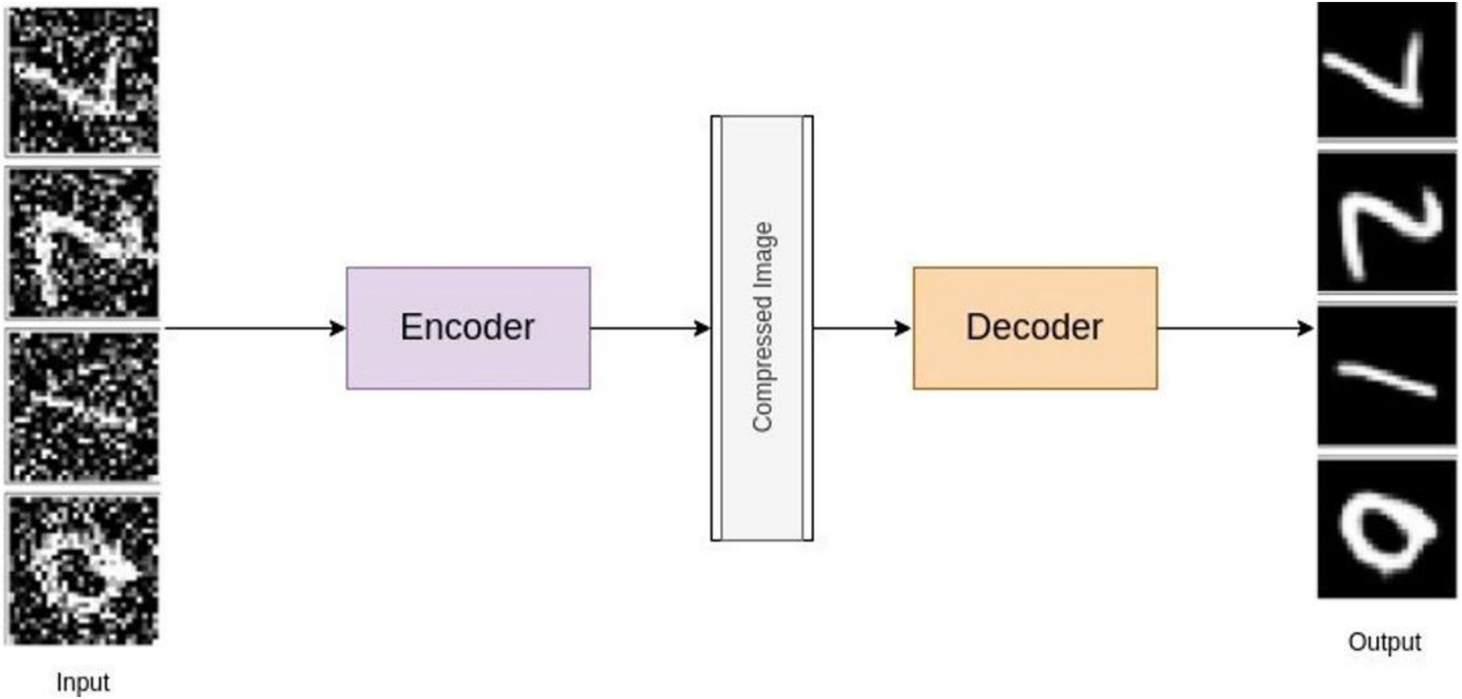
https://blog.keras.io/building-autoencoders-in-keras.html https://stackabuse.com/autoencoders-for-image-reconstruction-in-python-and-keras/

Больше – лучше!





Можно даже такое делать!

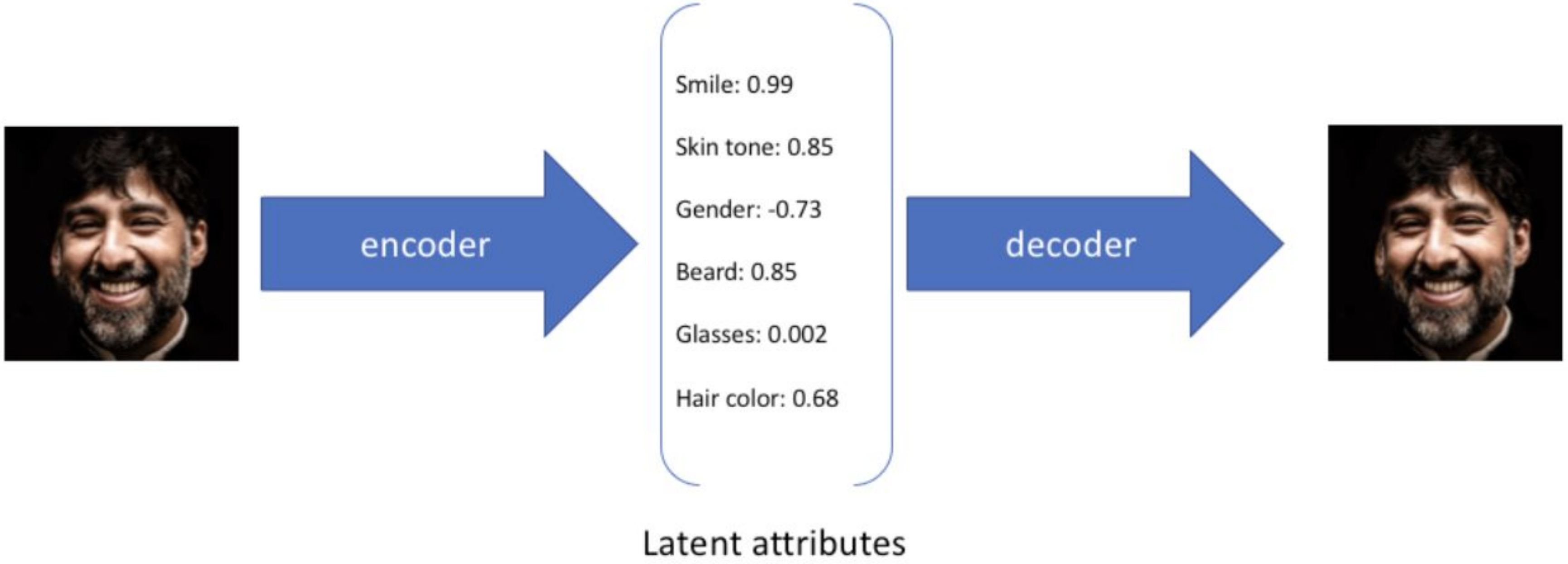


https://medium.com/@garimanishad/reconstruct-corrupted-data-using-denoising-autoencoder-python-code-aeaff4b0958e

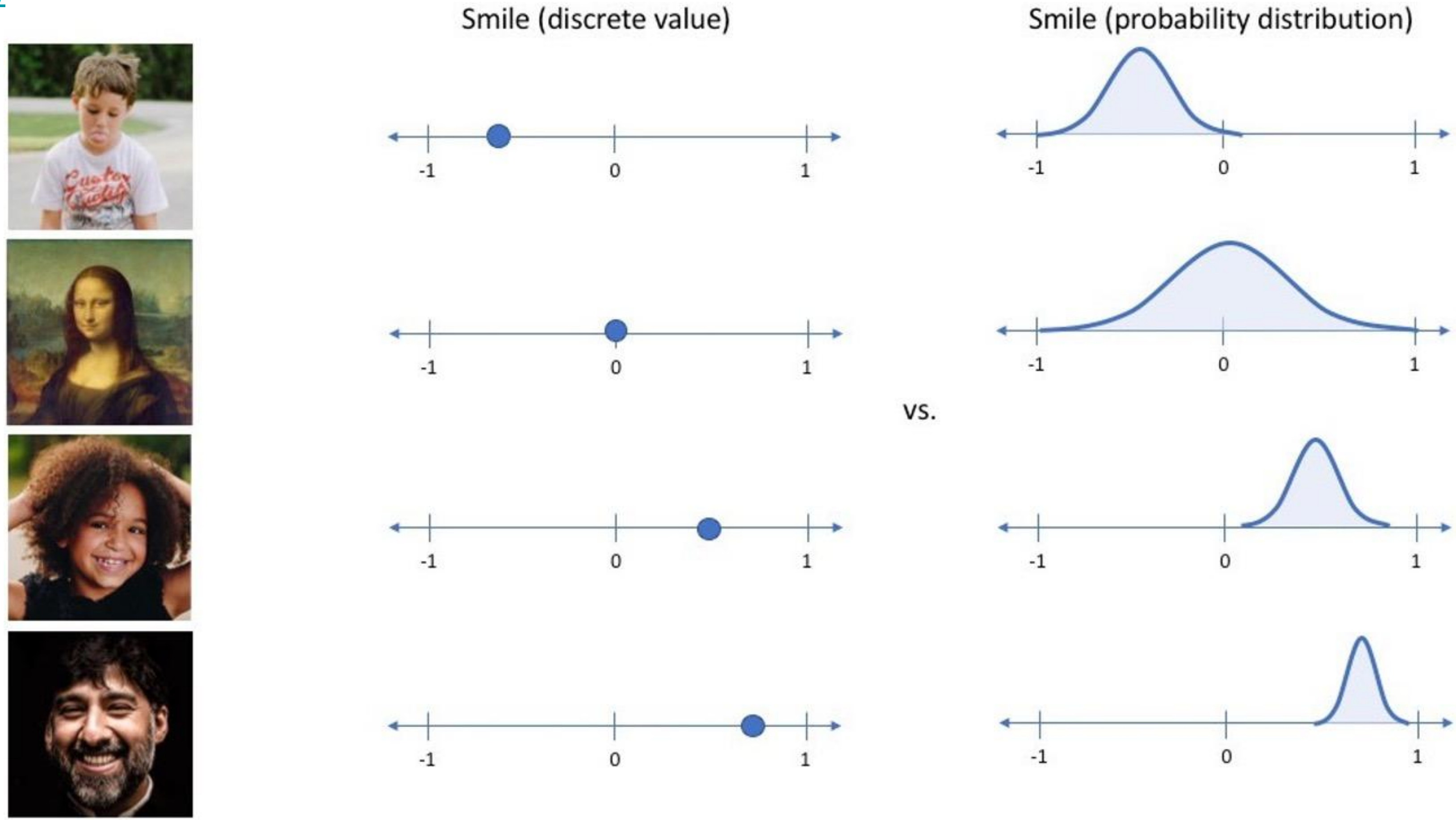
Давайте подумаем зачем оно нам такое надо…

https://www.jeremyjordan.me/variational-autoencoders/

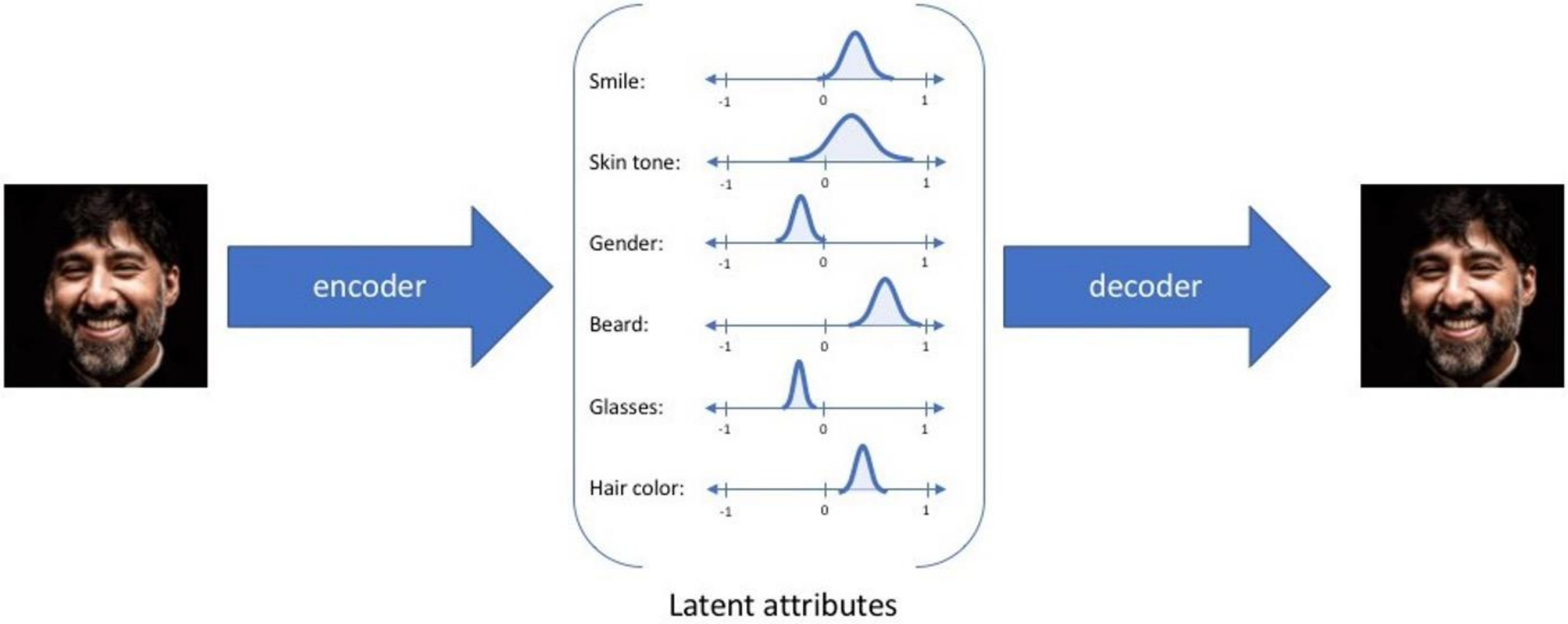
Variational AE = AE but better



https://www.jeremyjordan.me/variational-autoencoders/

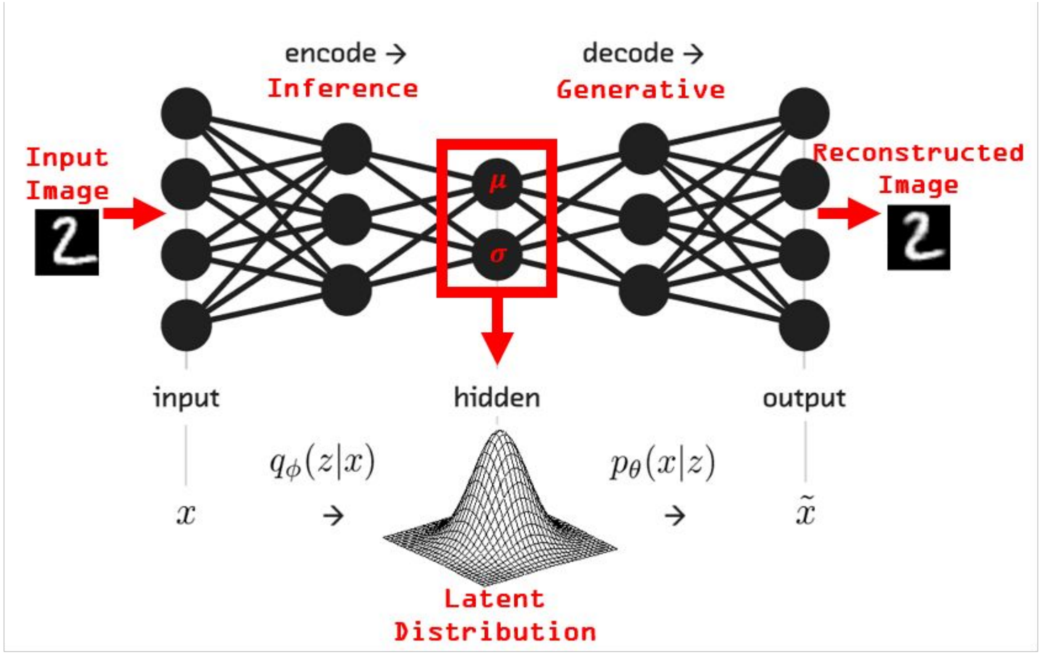


https://www.jeremyjordan.me/variational-autoencoders/



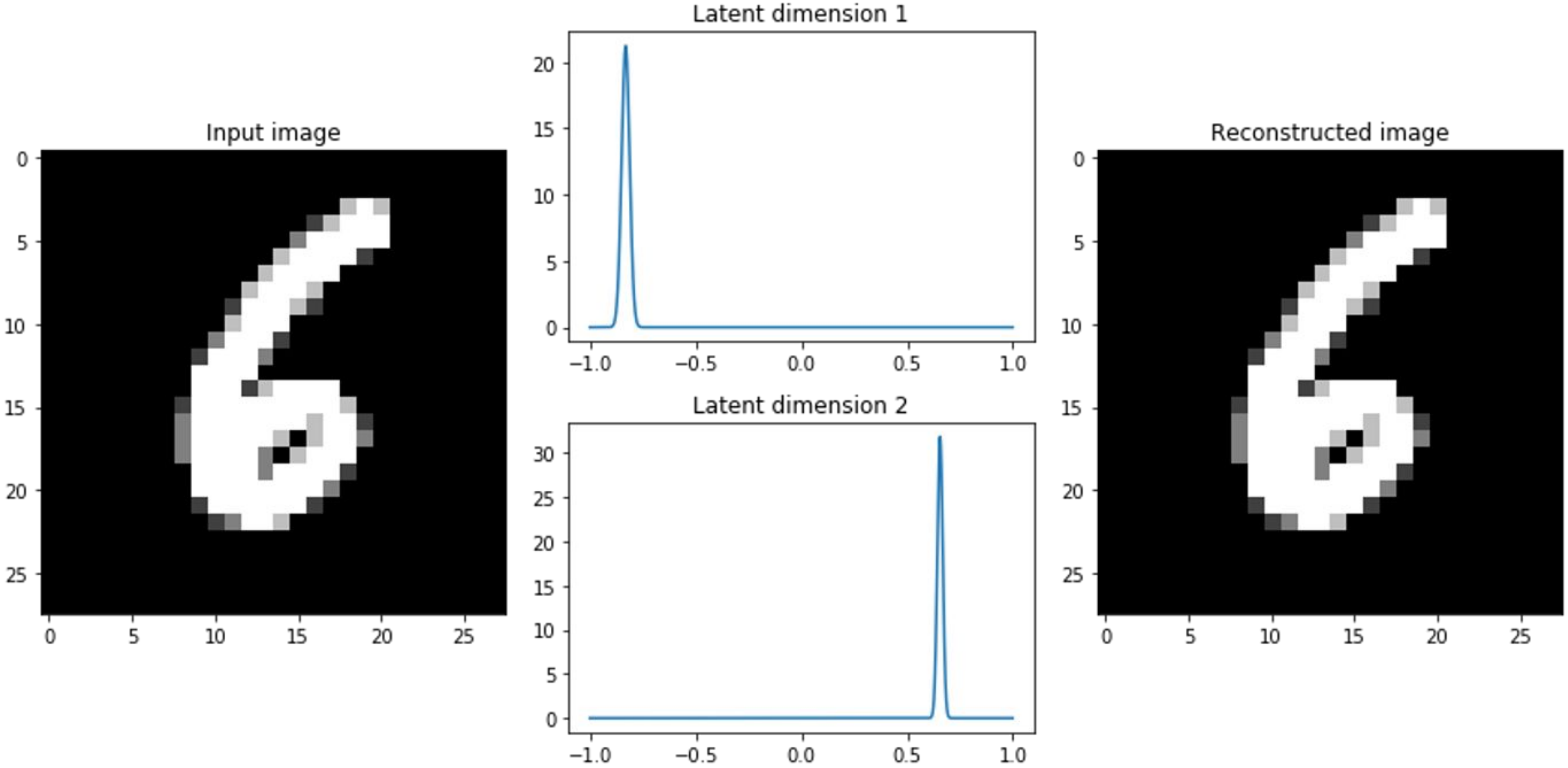
Фактически

$$\text{encoder}(x) = (\mu(x), \sigma(x))$$



https://arxiv.org/pdf/1712.08838

Но как мы не хотим



https://www.jeremyjordan.me/variational-autoencoders/

И как учить?

$$E_{q(z|x)} \log p(x|z) - KL(q(z|x)||p(z)) \rightarrow \max$$

Reconstruction Regularization

$$D_{\mathrm{KL}}(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)}.$$

Реконструкция обычно MSE

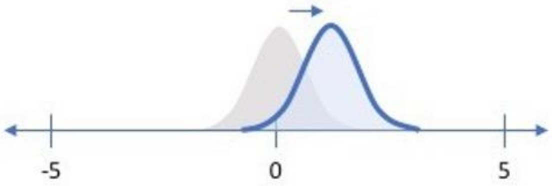
Если decoder выдает среднее гауссианы

$$p(x \mid z) = \mathcal{N}(x \mid \mu_\theta(z), \sigma^2 I)$$

$$\log p(x \mid z) = -\frac{1}{2\sigma^2}\|x - \mu_\theta(z)\|^2 - \frac{d}{2}\log(2\pi\sigma^2)$$

$$-\mathbb{E}_{q(z|x)}[\log p(x|z)] = \frac{1}{2\sigma^2}\mathbb{E}_{q(z|x)}\|x - \mu_\theta(z)\|^2 + \text{const}$$

Penalizing reconstruction loss encourages the distribution to describe the input



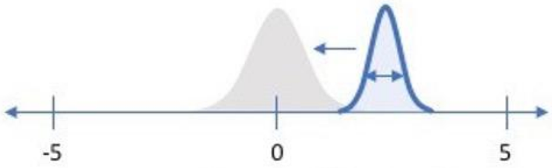
Our distribution deviates from the prior to describe some characteristic of the data

Without regularization, our network can “cheat” by learning narrow distributions

With a small enough variance, this distribution is effectively only representing a single value

Penalizing KL divergence acts as a regularizing force

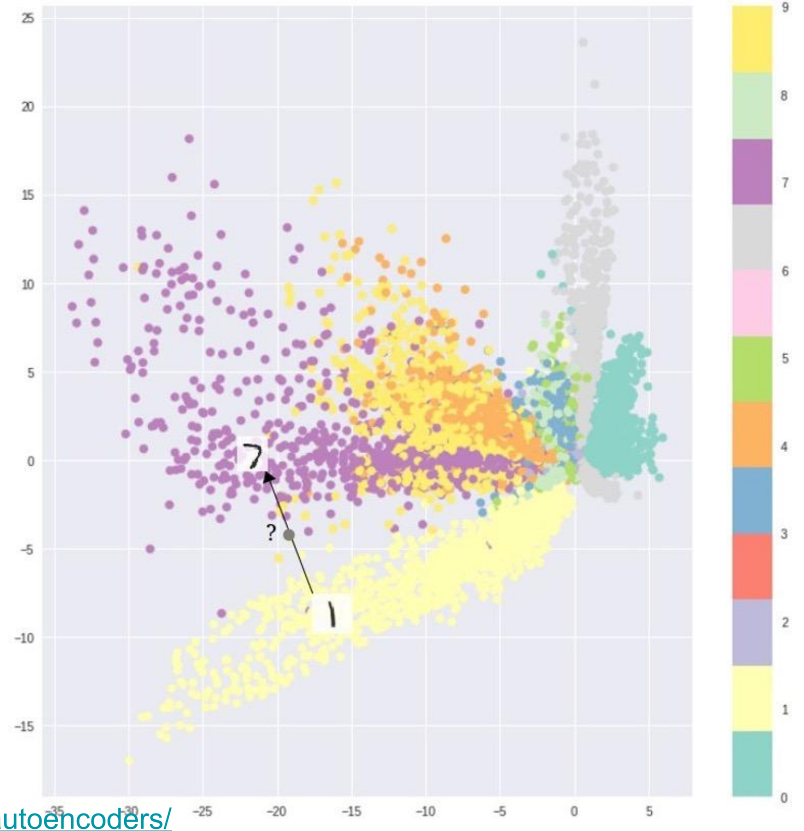
Attract distribution to have zero mean



Ensure sufficient variance to yield a smooth latent space

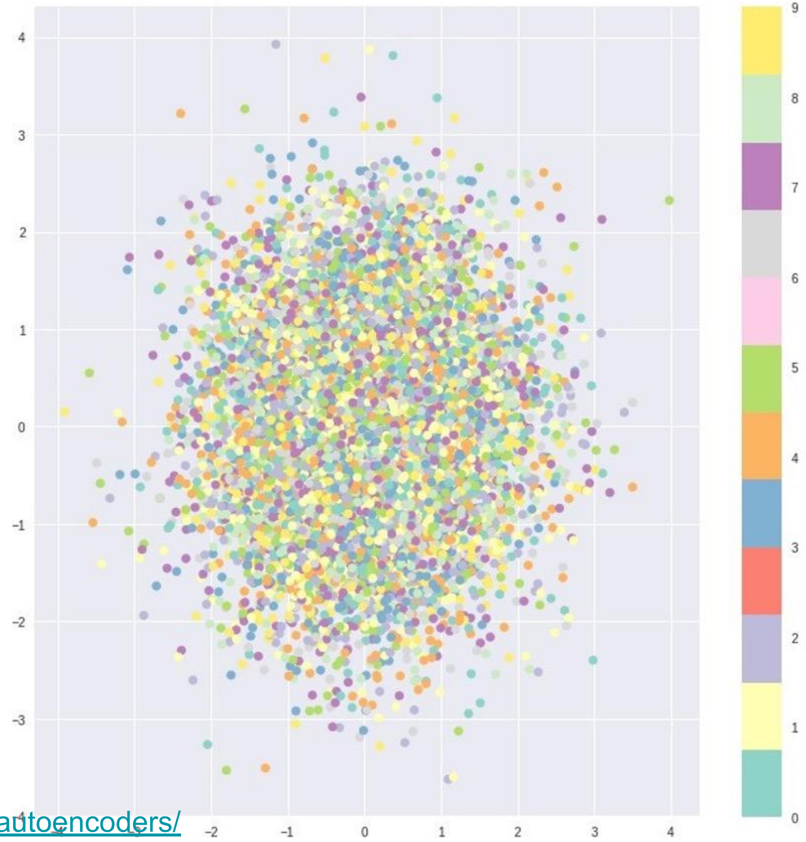
https://www.jeremyjordan.me/variational-autoencoders/

Только reconstruction



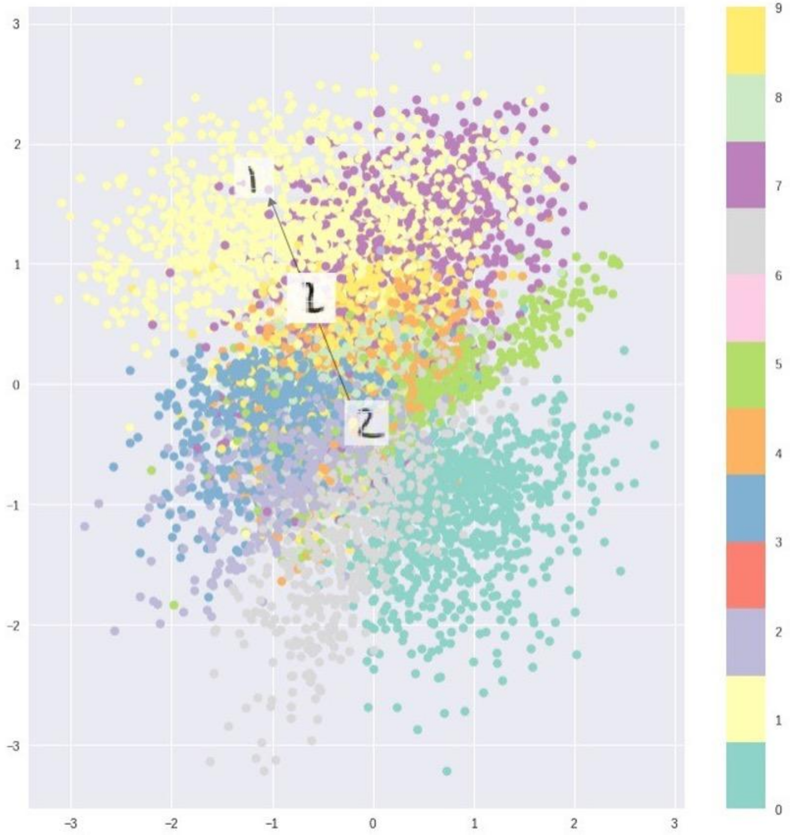
https://www.jeremyjordan.me/variational-autoencoders/

Только KL

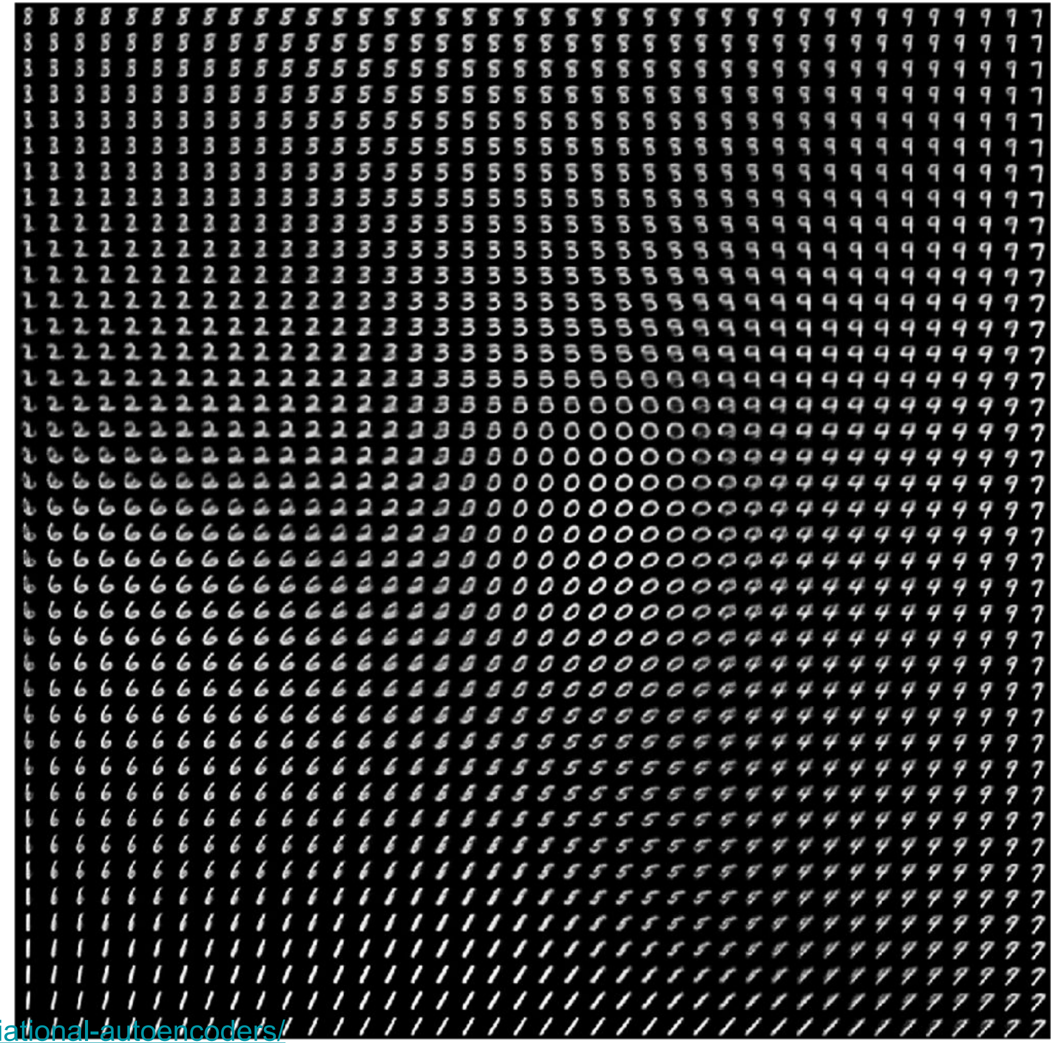


https://www.jeremyjordan.me/variational-autoencoders/

Оба

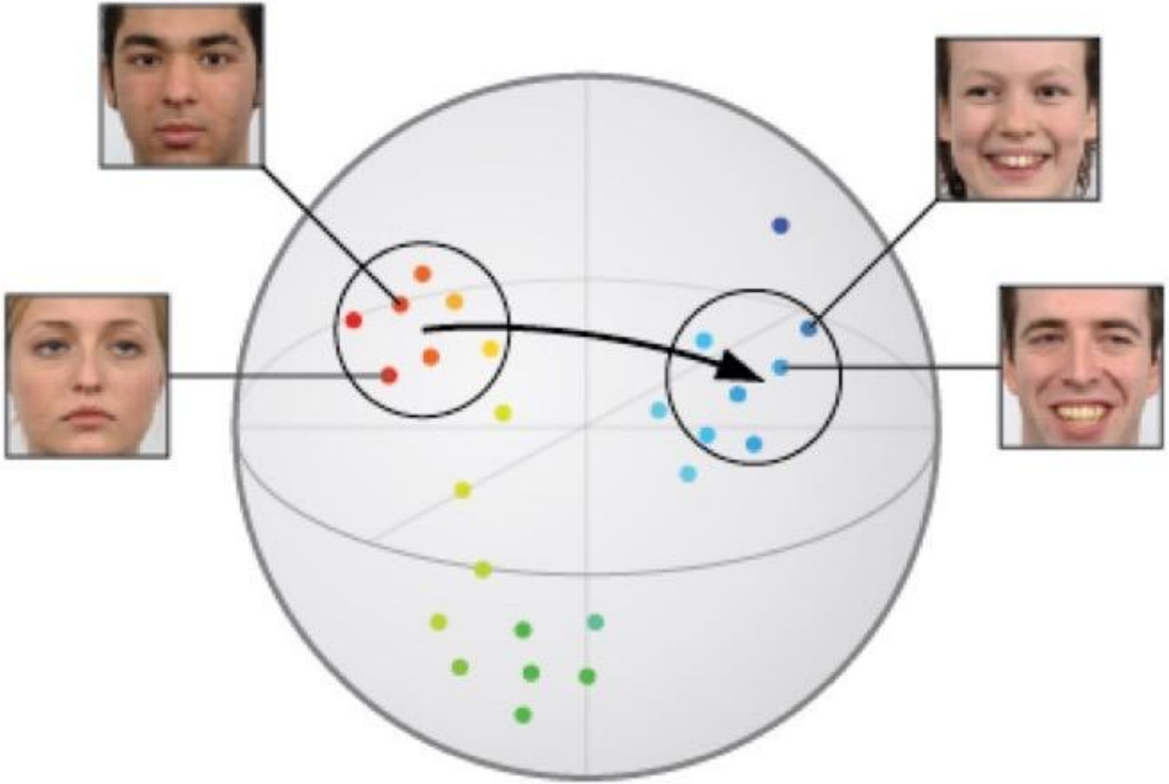


https://www.jeremyjordan.me/variational-autoencoders/



https://www.jeremyjordan.me/variational-autoencoders/

Улыбочку!



https://arxiv.org/pdf/1609.04468.pdf

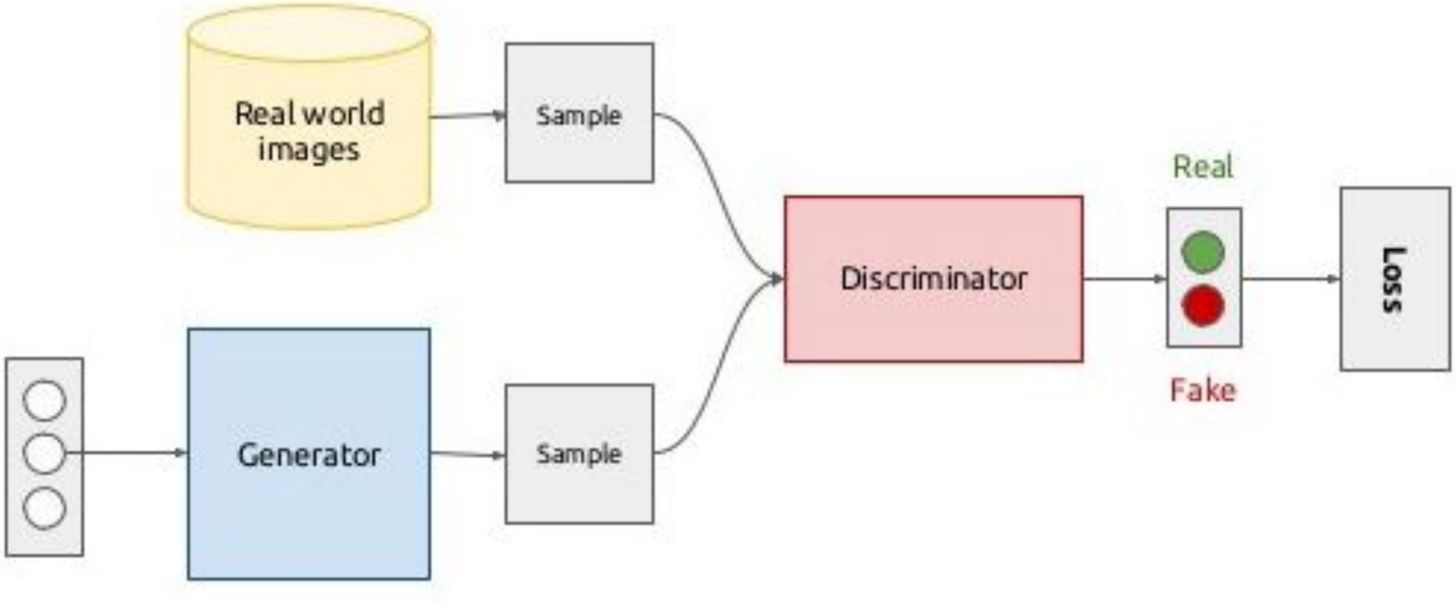
Reconstruction | Gender-biased smilevector | Gender-balanced smilevector

https://arxiv.org/pdf/1609.04468.pdf

Генерация

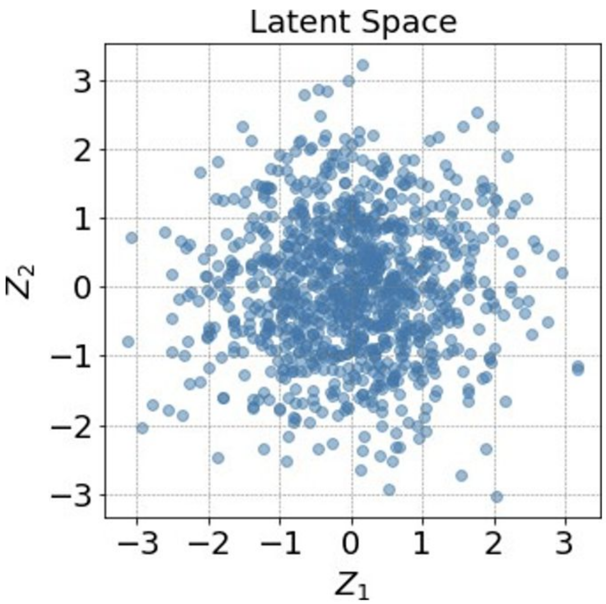
https://gaussian37.github.io/deep-learning-chollet-8-4/

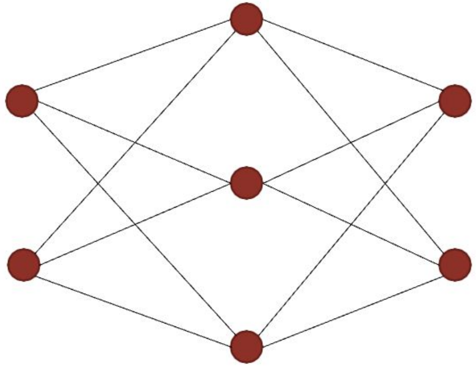
Новый тип модели! GAN

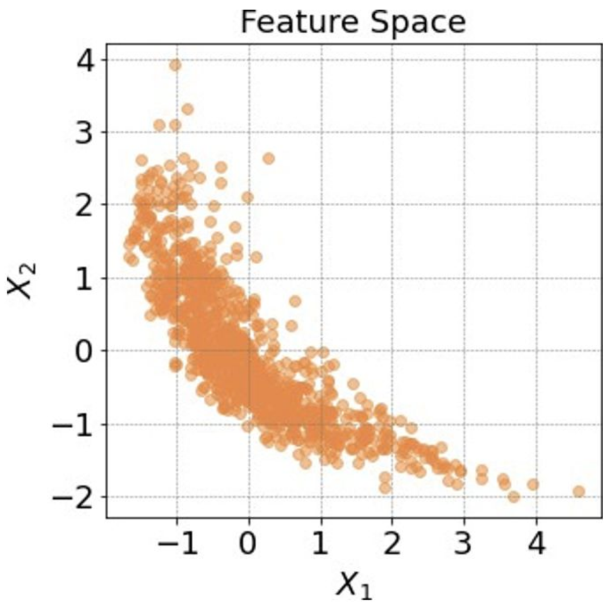


https://machine-learning.paperspace.com/wiki/generative-adversarial-network-gan

Generator



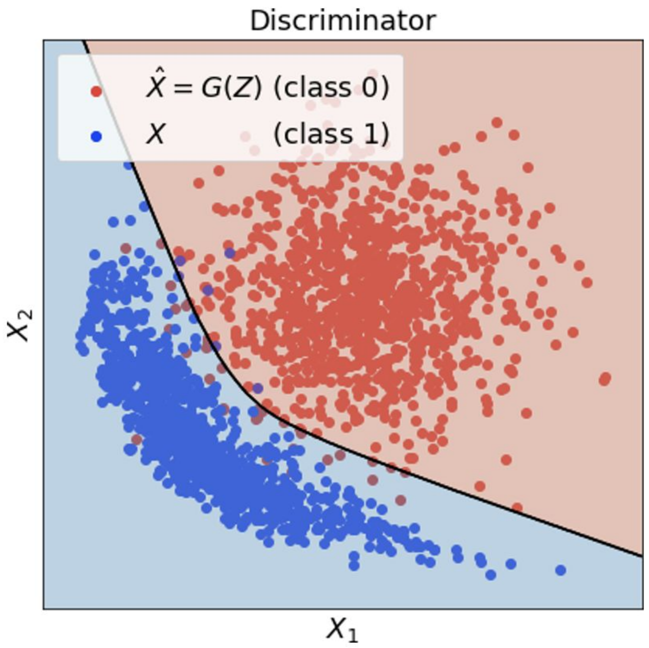




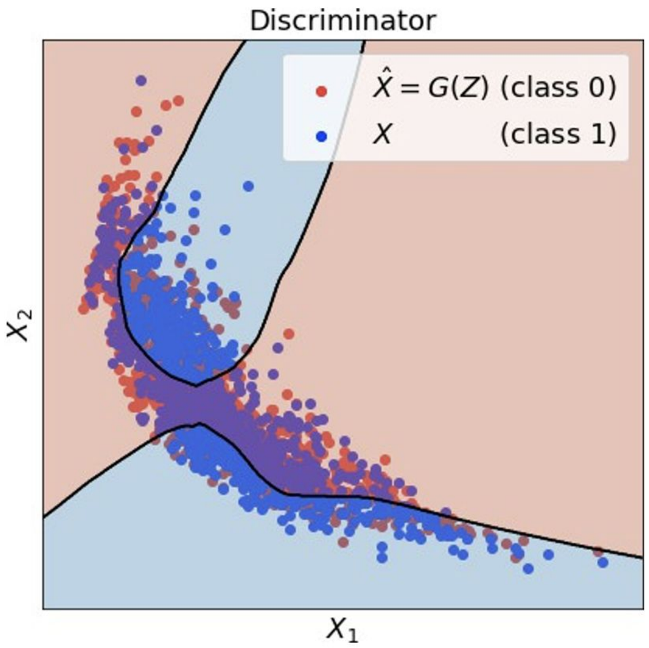
Z $\qquad$ $G(Z)$ $\qquad$ $X = G(Z)$

Credit: Elena Kantonistova

Discriminator



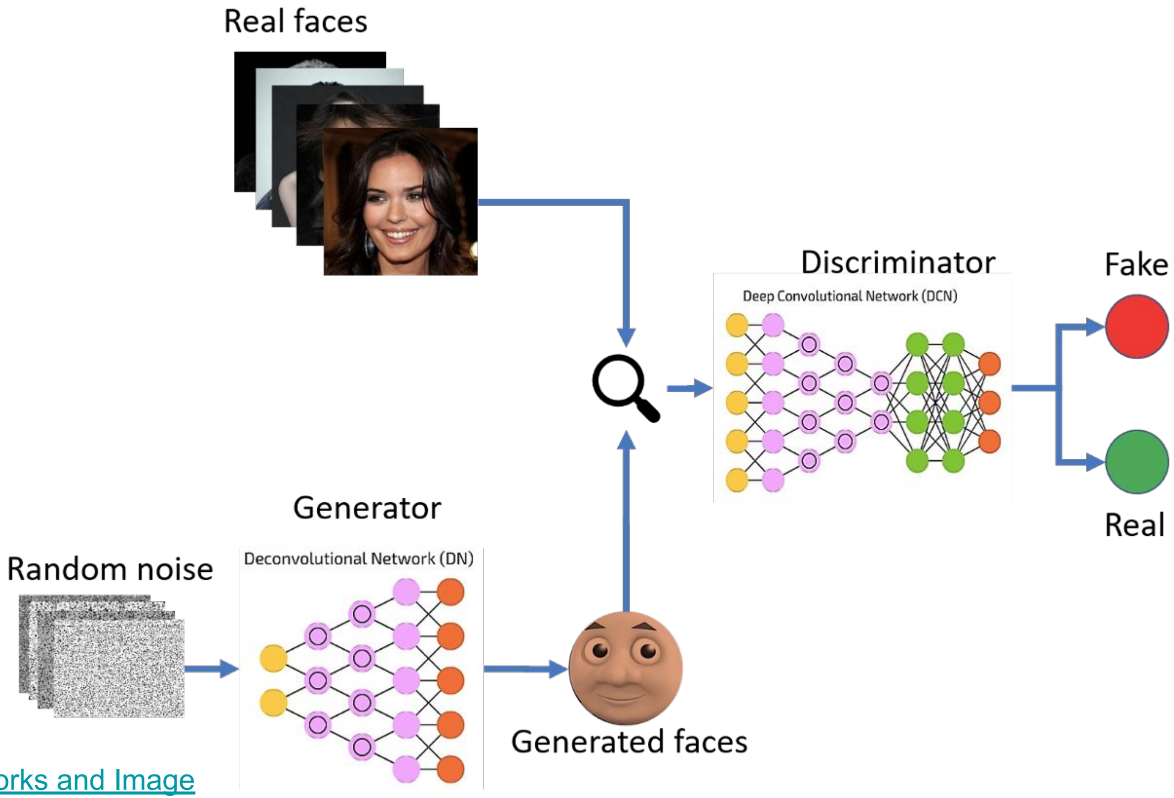
ROC AUC = 0.995
Log Loss = 0.135



ROC AUC = 0.554
Log Loss = 0.689

Credit: Elena Kantonistova

Суммарно



[Convolutional Neural Networks and Image Processing II ...]

Как учить?

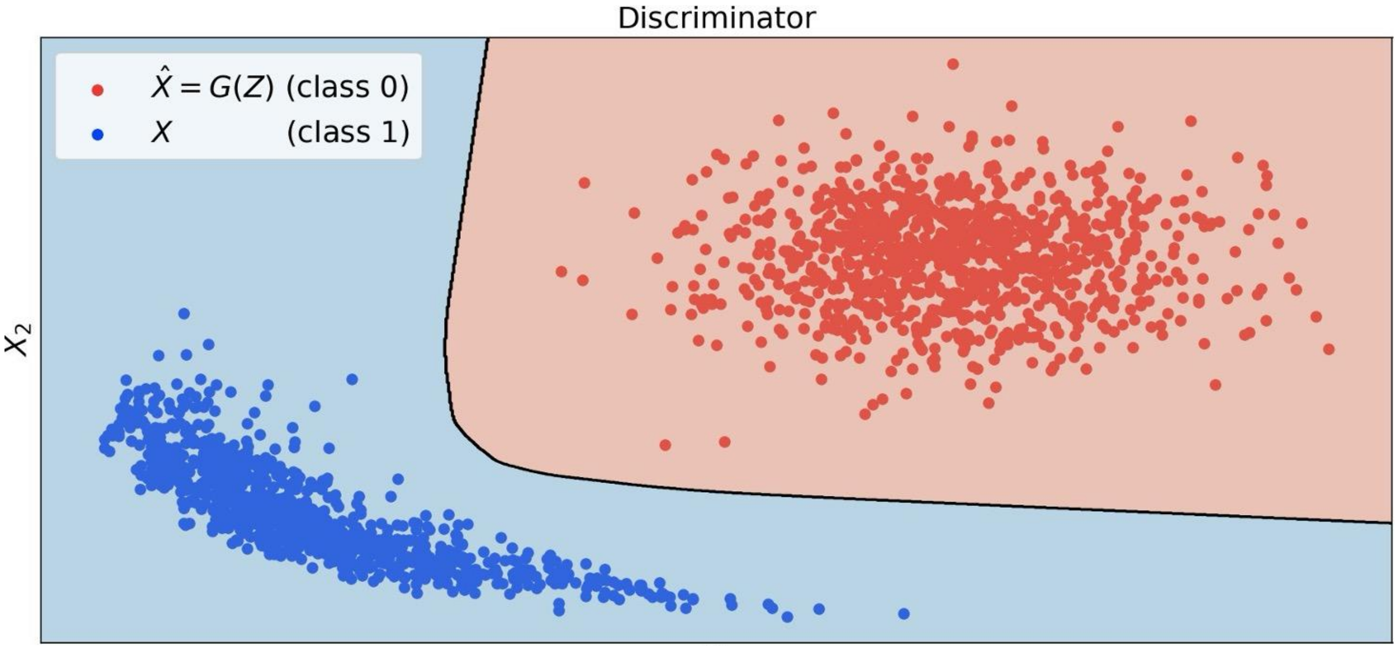
$$L(\boldsymbol{D}, \boldsymbol{G}) = -\frac{1}{n}\sum_{x_i \in X} \log \boldsymbol{D}(x_i) - \frac{1}{n}\sum_{z_i \in Z} \log\big(1 - \boldsymbol{D}(\boldsymbol{G}(z_i))\big)$$

$$\max_{\boldsymbol{G}} \min_{\boldsymbol{D}} L(\boldsymbol{D}, \boldsymbol{G})$$

Результаты

https://www.researchgate.net/figure/Progress-of-human-face-generation-with-GAN_fig2_338979046

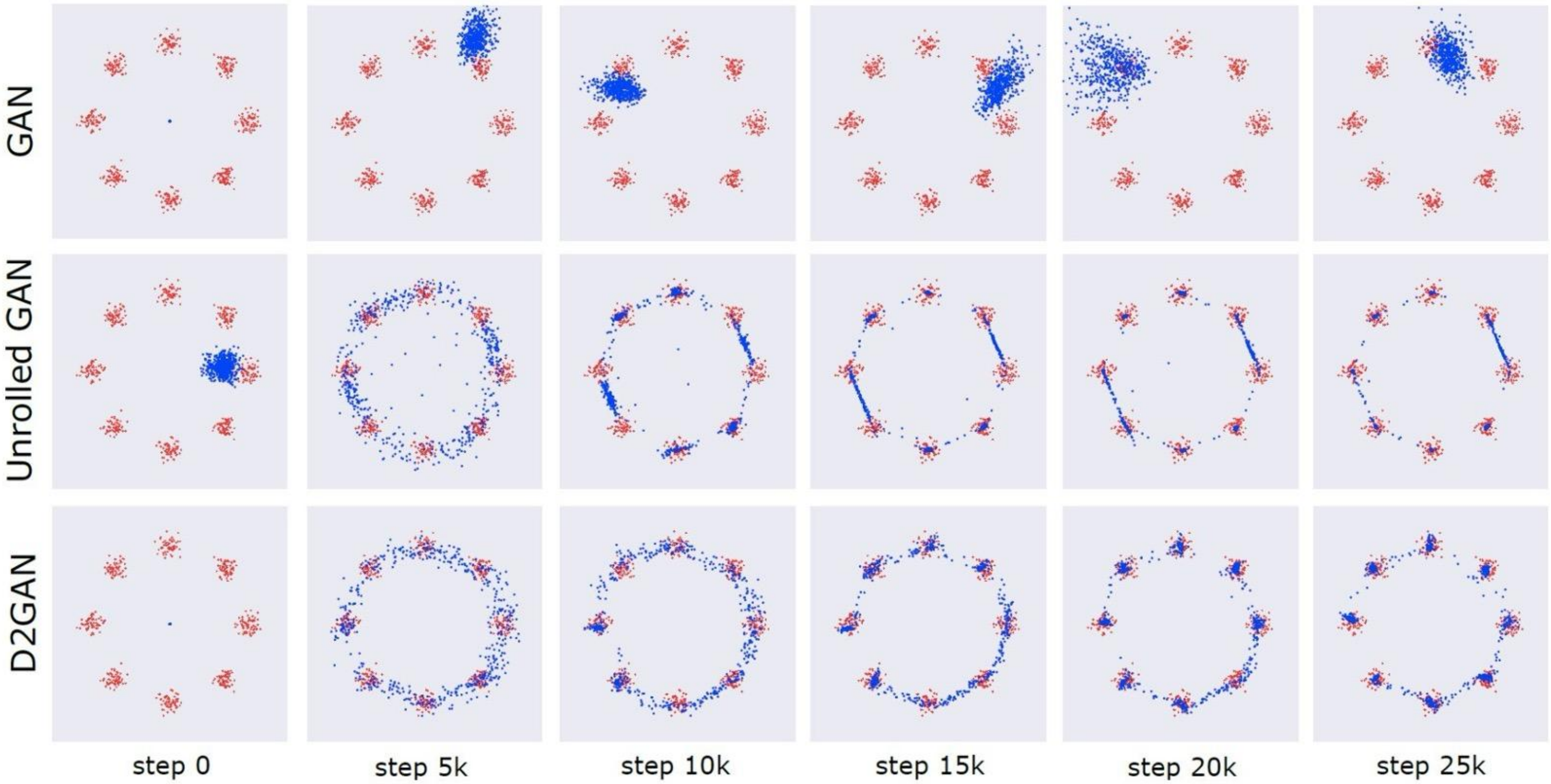
Проблемы



$$L(\boldsymbol{D}, \boldsymbol{G}) = -\frac{1}{n} \sum_{x_i \in X} \log \boldsymbol{D}(x_i) - \frac{1}{n} \sum_{z_i \in Z} \log\big(1 - \boldsymbol{D}(\boldsymbol{G}(z_i))\big)$$

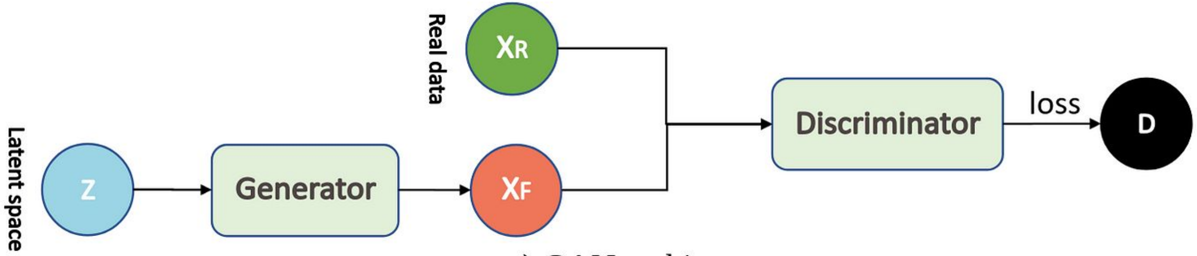
Credit: Elena Kantonistova

Схлопывание мод

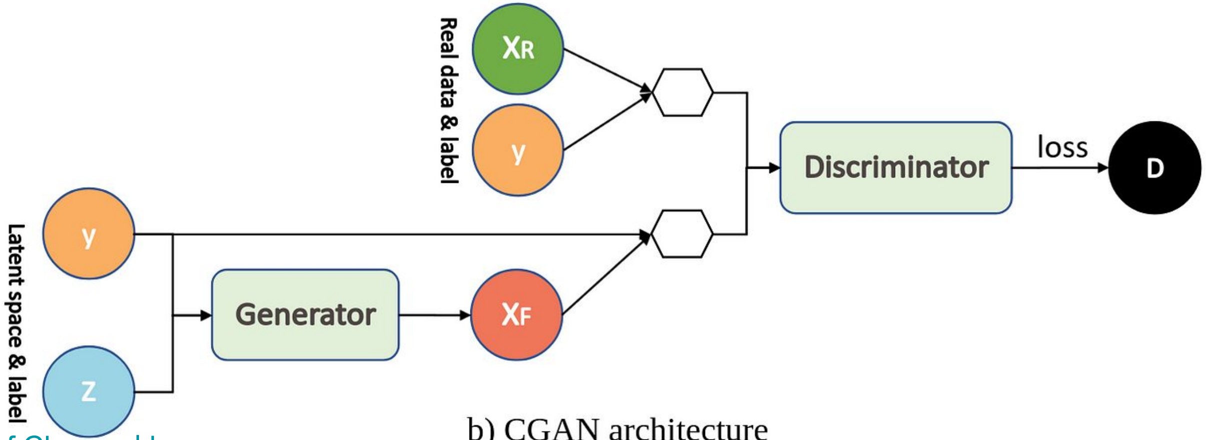


https://arxiv.org/pdf/1709.03831

Conditional GAN



a) GAN architecture



b) CGAN architecture

[Conditional GANs: Winds of Change | by California Youth STEM Project | Medium]

VAE Math

$$\mathbf{z} \sim p(\mathbf{z})$$

$$\mathbf{x} \sim p(\mathbf{x}|\mathbf{z})$$

$$\begin{aligned} p(\mathbf{x}) &= \int p(\mathbf{x}|\mathbf{z})\, p(\mathbf{z})\, \mathrm{d}\mathbf{z} \\ &= \mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}\left[p(\mathbf{x}|\mathbf{z})\right] \\ &\approx \frac{1}{K}\sum_{k} p(\mathbf{x}|\mathbf{z}_k) \end{aligned}$$

But $\mathbf{z} \in \mathbb{R}^M$

Oh no, we have a problem!

https://jmtomczak.github.io/blog/4/4_VAE.html

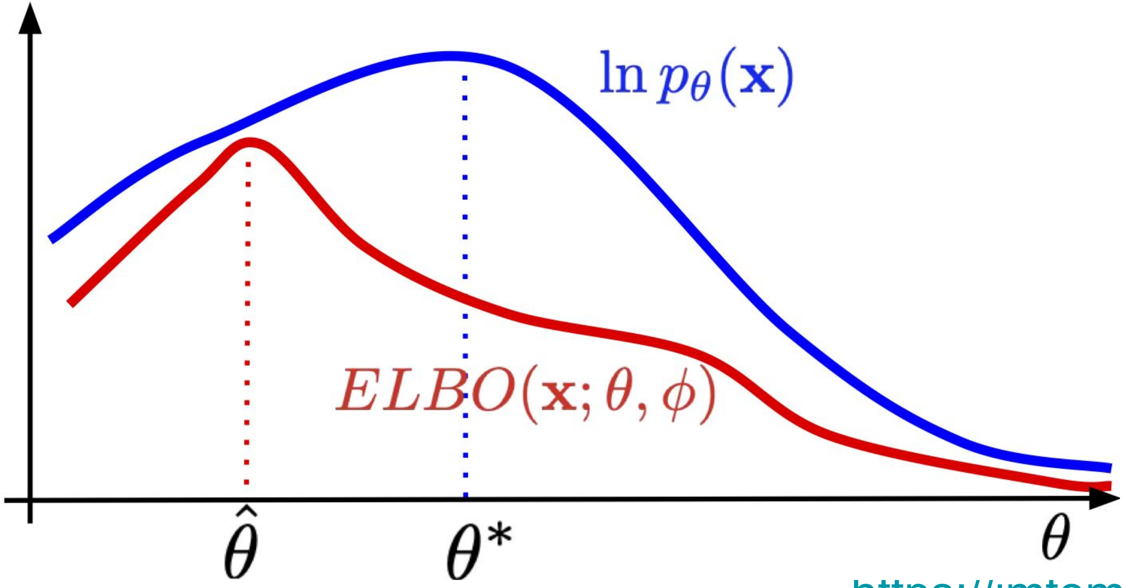
VAE

$$
\begin{aligned}
\ln p(\mathbf{x}) &= \mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}\left[\ln p(\mathbf{x})\right] \\
&= \mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}\left[\ln \frac{p(\mathbf{z}|\mathbf{x})p(\mathbf{x})}{p(\mathbf{z}|\mathbf{x})}\right] \\
&= \mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}\left[\ln \frac{p(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p(\mathbf{z}|\mathbf{x})}\right] \\
&= \mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}\left[\ln \frac{p(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p(\mathbf{z}|\mathbf{x})}\frac{q_\phi(\mathbf{z}|\mathbf{x})}{q_\phi(\mathbf{z}|\mathbf{x})}\right] \\
&= \mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}\left[\ln p(\mathbf{x}|\mathbf{z})\frac{p(\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\frac{q_\phi(\mathbf{z}|\mathbf{x})}{p(\mathbf{z}|\mathbf{x})}\right] \\
&= \mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}\left[\ln p(\mathbf{x}|\mathbf{z}) - \ln \frac{q_\phi(\mathbf{z}|\mathbf{x})}{p(\mathbf{z})} + \ln \frac{q_\phi(\mathbf{z}|\mathbf{x})}{p(\mathbf{z}|\mathbf{x})}\right] \\
&= \mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}\left[\ln p(\mathbf{x}|\mathbf{z})\right] - KL\left[q_\phi(\mathbf{z}|\mathbf{x})\|p(\mathbf{z})\right] + KL\left[q_\phi(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}|\mathbf{x})\right].
\end{aligned}
$$

https://jmtomczak.github.io/blog/4/4_VAE.html

VAE

$$\ln p(\mathbf{x}) = \underbrace{\mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}\left[\ln p(\mathbf{x}|\mathbf{z})\right] - KL\left[q_\phi(\mathbf{z}|\mathbf{x})\|p(\mathbf{z})\right]}_{ELBO} + \underbrace{\cancel{KL\left[q_\phi(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}|\mathbf{x})\right]}}_{\geq 0}$$



https://jmtomczak.github.io/blog/4/4_VAE.html

GAN Math

See https://arxiv.org/pdf/1406.2661